

Program-1

Create a collection named student having fields name, age, standard, percentage. Insert 5 to 10 random documents in collection.

- 1) Find name of all students having age>5
- 2) Increase the standard for all students by 1.
- 3) Arrange all the records in descending order of age.
- 4) Show the name of student who is the oldest student among all students.
- 5) Delete the record of the student if standard is 12.

```
db.student.insertMany([{"name":"abc",age:13,standard:6,perc:80},{name:"def",age:15,standard:8,perc:90},{name:"ghi",age:10,standard:3,perc:75},{name:"pqr",age:5,standard:1,perc:89},{name:"xyz",age:17,standard:12,perc:97}])
```

- 1) `db.student.find({age:{$gt:5}},{name:1,_id:0})`
- 2) `db.student.updateMany({},{$inc:{standard:1}})`
- 3) `db.student.find().sort({age:-1})`
- 4) `db.student.find({}, {name:1, _id:0}).sort({age:-1}).limit(1)`
- 5) `db.student.deleteOne({standard:12})`

Program-2.

Perform the tasks as asked below.

Create Collection “employees” with following data

```
[{_id: 1,name: "Eric",age: 30,position: "Full Stack Developer",salary: 60000},
{_id: 2,name: "Erica",age: 35,position: "Intern",salary: 8000},
{_id: 3,name: "Erica",age: 40,position: "UX/UI Designer",salary: 56000},
{_id: 4,name: "treric7",age: 37,position: "Team Leader",salary: 85000},
{_id: 5,name: "Eliza",age: 25,position: "Software Developer",salary: 45000},
{_id: 6,name: "Trian",age: 29,position: "Data Scientist",salary: 75000},
{_id: 7,name: "Elizan",age: 25,position: "Full Stack Developer",salary: 49000}]
```

- 1) Find All Documents
- 2) Find Documents by Position “Full Stack Developer”:
- 3) Retrieve name of employees whose age is greater than or equal to 25 and less than or equal to 40.
- 4) Retrieve name of the employee with the highest salary.
- 5) Retrieve employees with a salary greater than 50000.
- 6) Retrieve employees' names and positions, excluding the “_id” field.
- 7) Count the number of employees who have salary greater than 50000
- 8) Retrieve employees who are either “ **Software Developer**” or “**Full Stack Developer**” and are below 30 years.
- 9) Increase the salary of an employee who has salary less than 50000 by 10%.
- 10) Delete all employees who are older than 50.
- 11) Give a 5% salary raise to all “**Data Scientist**”
- 12) Find documents where name like “%an”

- 13) Find documents where name like “Eri--” (Case Insensitive)
- 14) Find documents where name like “%ric%”
- 15) Find documents where name contains only 4 or 5 letters.
- 16) Find documents where name must end with digit

Program-3

Insert 10 documents with random data with fields _id,brand,price,cat as shown below.

```
db.product.insertMany([
  { _id:1,brand:"samsung",price:29000,cat:"mobile"},
  { _id:2,brand:"nokia",price:5000,cat:"mobile"},
  { _id:3,brand:"vivo",price:16000,cat:"mobile"},
  { _id:4,brand:"samsung",price:60000,cat:"tv"},
  { _id:5,brand:"samsung",price:40000,cat:"washing machine"},
  { _id:6,brand:"ifb",price:45000,cat:"wasing machine"},
  { _id:7,brand:"apple",price:120000,cat:"mobile"},
  { _id:8,brand:"oppo",price:20000,cat:"mobile"},
  { _id:9,brand:"sony",price:80000,cat:"tv"},
  { _id:10,brand:"vivo",price:31000,cat:"mobile"},
])
```

- 1) Display price and brand of product which are of mobile cat.**
- 2) Increase price of each Samsung products by 1000.**
- 3) Update all vivo product by adding field quantity and add random value**
- 4) Display price of products which are of vivo or oppo brand.**
- 5) Display brand and cat of products which are less than 80000 and greater than or equal to 30000.**